

# Supplementary Material: Repair-Aware Quantum Sampling for a Discrete Process-Design QUBO

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## ABSTRACT

This supplement records provenance, validation, and artifact anchors for the DS-MFG repair-aware quantum-sampling manuscript. It is intended to be compiled and submitted separately from the main conference paper if appendices are not counted as article body material in the submission workflow.

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Keywords: supplementary material, quantum optimization, process design, reproducibility

## REPOSITORY AND PACKAGING

The main manuscript keeps only the narrative-critical values. This supplement keeps the audit data close to the manuscript source while allowing the main article to remain within the conference page limit. The repository URL is <https://github.com/SECQUOIA/pse-quantum-discrete-optimization>. A final archival DOI has not yet been assigned. It will be added before camera-ready submission if one is minted; the review artifact is identified by the repository snapshot that contains this supplement and by the checksums below.

Table 1: Current reproducibility pins validated for the DS-MFG case study.

| Layer                      | Current pin or check                                                                                                      | Repository anchor                                                                                  |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Julia optimization wrapper | QiskitOpt.jl v0.7.0; manifest git tree a601dc cdd43fd0a8b18e63f3b771273817af49eb.                                         | case-studies/ds-mfg-qubo-qiskitopt/Project.toml; case-studies/ds-mfg-qubo-qiskitopt/Manifest.toml. |
| Julia QUBO stack           | QUBODrivers.jl v0.6.1, QUBOTools.jl v0.13.0, QUBO.jl v0.6.0.                                                              | case-studies/ds-mfg-qubo-qiskitopt/Project.toml; case-studies/ds-mfg-qubo-qiskitopt/Manifest.toml. |
| Python bridge              | Qiskit 2.3.1, Qiskit Aer 0.17.2, Qiskit Optimization 0.7.0, Qiskit IBM Runtime 0.46.1, NumPy 2.2.6, SciPy 1.15.3.         | case-studies/ds-mfg-qubo-qiskitopt/CondaPkg.toml.                                                  |
| Fresh-clone validation     | julia --project=. scripts/smoke_test.jl imports packages, checks required artifacts, and validates cached summary values. | case-studies/ds-mfg-qubo-qiskitopt/scripts/smoke_test.jl.                                          |

Table 2: Reproducibility levels used for the DS-MFG manuscript package.

| Level                            | Meaning                                                                                                                                                                 | Current support                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Score-from-cache reproducible    | Retained samples, summaries, and repair tables can be inspected and rescored without rerunning the stochastic campaigns.                                                | Supported by cached distributions, hit-rate reports, checksums, and the smoke test. This is the main reproducibility level for the manuscript claims.                                                                                                                                                                                              |
| Rerunnable with retained scripts | The scripts and pinned environment can rerun selected QAOA/VQE, classical, and hardware-handoff workflows, subject to stochastic variation and credential availability. | Partially supported. QAOA fixed-angle parameters and sampled outputs are retained; selected VQE metadata reruns now retain optimized ansatz vectors for seed 74018 and for the final-budget selected seeds 74018, 74007, and 74001. Those metadata reruns are provenance artifacts, not replacements for the historical sampled VQE distributions. |
| Full campaign replay             | Every stochastic campaign can be reproduced byte-for-byte from retained optimizer states, simulator logs, hardware calibration, and submitted-circuit metadata.         | Not claimed. Some optimizer, MPS truncation, and submitted-job hardware metadata were not retained.                                                                                                                                                                                                                                                |

Table 3: Checksums for retained archives and primary cached summaries.

| Artifact                                                                                                                      | SHA-256 checksum                                                 |
|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| case-studies/ds-mfg-qubo-qiskitopt/Fw_DSsmfgcasequboinformation.zip                                                           | aa20d793fc45c25c2bd7cc0a231877f71b225bf80daee1531cef6eb6cac5af2  |
| ds_mfg_ibm_qaoa_pilot_fez_4096x3.zip                                                                                          | 88af6f74b0fc10dcece69e783c37782649d1082e3220a88c61f389274665dff9 |
| ds_mfg_direct_full_qubo_hardware_pilot.zip                                                                                    | fe7a60bd6576e66da9059d8aa6831424ef6fcbbf47bd934e60850b7933558091 |
| case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_qaoa_juliquao_transfer_highread/qaoa_juliquao_transfer_summary.csv                  | 10bb3bc9b746fb12a87eadc1fd2b819c1a3cacbc84025c9bd54797c1088c34b7 |
| case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_vqe_reduced_flow_objective_final/vqe_reduced_top50_sampling_summary.csv             | aac0299d49314b8d0e4d54b088f50cc728f1d212540613348ab62d209a132ee1 |
| case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_gurobi_provenance/ip_exact_enumeration_summary.json                                 | 1298dcb07aa82c0adc05ead3fb1b3f4b86ae9ef38e9a276b98f97ec5a7606e5e |
| case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_gurobi_provenance/gurobi_local_pool_rerun_summary.json                              | 13f7683c41cbf14b4225326e41778c6f42ae3979e4d1303af65c7132fe35a6e7 |
| case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_vqe_reduced_flow_objective_selected_metadata/vqe_reduced_top50_sampling_summary.csv | df82de49b921e3b70d5d24f0896e77b6da64477c3a13cfc0a0761d6e9ebd2db  |
| case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_hit_rate_reports/time_to_solution_report.csv                                        | c27e70684f3b753efc4eb8cf2b75e79afe4b15402bcd6b3ca7d7a598310f2efd |

## VALIDATION FACTS

Table 4: Numerical validation facts used to support the manuscript claims.

| Check                           | Retained value                                                                                                                                                                                                                                                                                                         | Anchor                                                                                                                                       |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Exact repaired optimum          | $F(y^*) = 11.7095$ at flow 1001110100111100011, ordered as $(f_{00}, \dots, f_{18})$ .                                                                                                                                                                                                                                 | ds_mfg_reduced_flow_objective/reduced_flow_summary.csv; ds_mfg_reduced_flow_objective/reduced_exact_top_flows.csv.                           |
| Gurobi pool bridge              | All 36 stored Gurobi-pool flows match the repaired QUBO score within $3.1 \times 10^{-13}$ .                                                                                                                                                                                                                           | MANUSCRIPT_FINDINGS.tex; Fw_DSsmfgcasequboinformation.zip.                                                                                   |
| Original Gurobi export metadata | Model capex: 19 binary variables, 20 constraints, 58 nonzeros; PoolSearchMode=2, PoolSolutions=36, solve time 0.073 s, 36 solutions found. Solver version, status code, MIPGap, PoolGap, and pool completeness/diversity intent were not retained in the original export.                                              | Fw_DSsmfgcasequboinformation.zip/gurobi_pool_solutions_n36_20260126_105706.json.                                                             |
| Local Gurobi rerun validation   | Gurobi 13.0.2 rerun of the same IP with PoolSearchMode=2 and PoolSolutions=100: OPTIMAL, MIPGap=0.0, best objective 11.7095, objective bound 11.7095, 36 solutions, local solve time 0.0077 s, and exact retained-pool agreement. PoolGap is null in the rerun summary and pool completeness/diversity is not claimed. | ds_mfg_gurobi_provenance/gurobi_local_pool_rerun_summary.json; ds_mfg_gurobi_provenance/gurobi_local_pool_rerun_solutions.csv.               |
| Quadratic surrogate weakness    | Global fit $R^2 \approx 0.999$ , but the surrogate optimum repairs to 14.5505. The true repaired optimum is surrogate rank 33; the surrogate optimum is exact repaired rank 6.                                                                                                                                         | ds_mfg_reduced_flow_objective/reduced_flow_summary.csv; Figure 1.                                                                            |
| Low-energy surrogate audit      | The exact top-50 repaired flows include 14 non-pool flows. In the lowest 1% repaired-flow set, exact-rank versus surrogate-rank Spearman correlation is 0.901. Exact enumeration wall time was not retained.                                                                                                           | ds_mfg_reduced_flow_objective/reduced_exact_top_flows.csv; MANUSCRIPT_FINDINGS.tex.                                                          |
| Primary QAOA simulator check    | JuliQAOA statevector probability on the reference optimum was $2.63 \times 10^{-3}$ for the energy-targeted $p = 5$ angles. The cached Aer MPS sampled rate was $2.53 \times 10^{-3}$ , inside its Wilson 95% interval $[2.35, 2.73] \times 10^{-3}$ . MPS truncation logs were not retained.                          | ds_mfg_qaoa_juliquaoa_angle_search/juliquaoa_angle_summary.csv; ds_mfg_qaoa_juliquaoa_transfer_highread/qaoa_juliquaoa_transfer_summary.csv. |

Table 5: Implications of missing provenance values for the manuscript claims.

| Missing value                                         | Affects main claim?                                                                                                                     | Affects full rerun?                                                                                          | Current mitigation                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Original-export Gurobi final status and MIPGap        | No for the current claim after rerun; the original export still lacks these fields, but the model/pool behavior has been rerun locally. | Partly; exact historical solver state is unavailable, but current solver provenance is retained.             | The same flow is certified by exact enumeration over all $2^{19}$ flows and by a local Gurobi 13.0.2 rerun with OPTIMAL status, MIPGap=0.0, 36 solutions, and exact retained-pool agreement. PoolGap and original pool-generation intent remain unclaimed.                                                        |
| MPS truncation logs                                   | No for the central QAOA sampling claim.                                                                                                 | Yes, for simulator-forensics replay.                                                                         | The energy-targeted QAOA row is cross-checked against independent JuliQAOA statevector probabilities, which agree with the Aer MPS sampled rate within the Wilson interval.                                                                                                                                       |
| VQE optimized parameter vectors                       | No; VQE is secondary and reported as a retained sampled distribution.                                                                   | Partly; selected fixed-circuit replay is now supported, but full historical optimizer replay is not claimed. | Metadata reruns retain optimized EfficientSU2 vectors for seed 74018 and for seeds 74018, 74007, and 74001 at the final sampling budget. The current metadata rerun produced 4 reference hits total and is provenance, not a replacement for the historical sampled VQE distributions reported in the manuscript. |
| Submitted hardware layout, depth, and two-qubit count | No, because the hardware row is a submission-and-scoring handoff only.                                                                  | Yes, for hardware-performance diagnosis.                                                                     | The manifest retains backend, job IDs, shots, seeds, logical circuit metadata, and scored counts. No noise-mechanism claim is made.                                                                                                                                                                               |
| Hardware calibration or mitigation metadata           | No under the current claim.                                                                                                             | Yes, for calibrated hardware analysis.                                                                       | The manuscript states that no readout or error mitigation is claimed and that the retained metadata cannot diagnose a specific noise source.                                                                                                                                                                      |
| Exact enumeration wall time                           | No.                                                                                                                                     | Partly, for cost accounting.                                                                                 | The enumeration is documented as exact over all $2^{19}$ flows; issue 24 tracks recording wall time if the enumeration script is rerun.                                                                                                                                                                           |

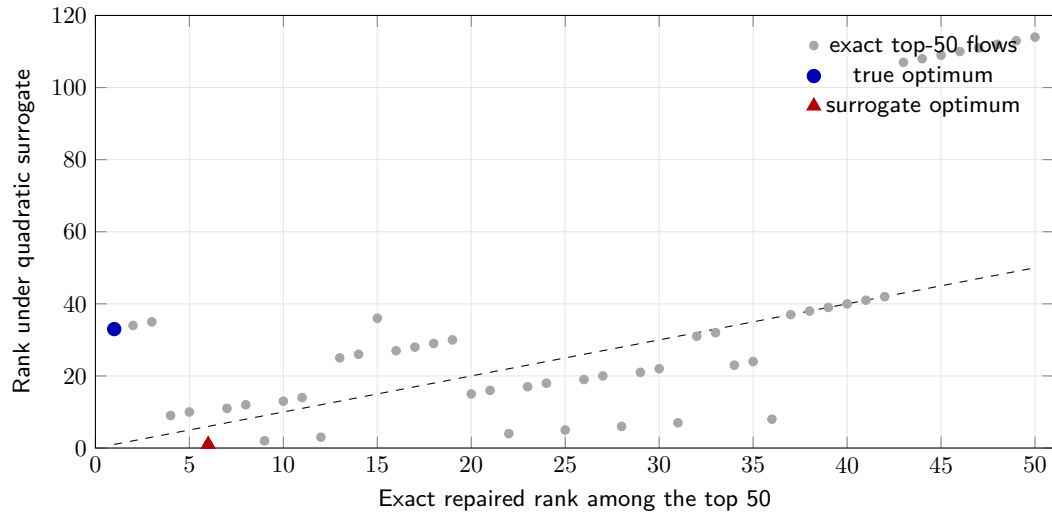


Figure 1. Surrogate rank audit for the exact top-50 repaired flows. The quadratic surrogate has high global  $R^2$  but imperfect low-energy ordering: the exact optimum is only rank 33 under the surrogate, while the surrogate minimum repairs to the sixth-best exact flow. Quantum-sampling outcomes are therefore scored using the exact repaired objective  $F$ , not the surrogate  $\hat{F}$ .

## VARIATIONAL AND HARDWARE PROVE- NANCE

Table 6: Variational-parameter provenance for the central QAOA and VQE rows.

| Workflow                                | Circuit / ansatz                                                             | Objective and optimizer                                                                                                                                  | Parameter source                                                                                                                                                                                  | Validation metric                                                                                                                                                                           |
|-----------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Energy-targeted QAOA                    | $p = 5$ QAOA on the 19-flow surrogate.                                       | Minimize $\mathbb{E}_{p_\theta}[\hat{F}(y)]$ with <code>JuliQAOA.find_angles_bh</code> ; seed 91001; 5 basin-hopping iterations.                         | <code>ds_mfg_qaoa_juliqaoa_angle_search/juliqaoa_angle_summary.csv</code> ; angles transferred beta-then-gamma to <code>QiskitOpt.QAOA</code> .                                                   | 664 reference hits / 262144 reads; 39661 top-50 reads.                                                                                                                                      |
| Top-10 QAOA diagnostic                  | $p = 5$ QAOA on the 19-flow surrogate.                                       | Maximize probability on the enumerated exact top-10 set; oracle-aided because the repaired ranking is known.                                             | Top-10 angle vector retained in <code>ds_mfg_ibm_qaoa_pi_lot_fez_4096x3.zip/job_manifest.json</code> ; local 1007-hit diagnostic CSV was not retained.                                            | 1007 reference hits / 262144 reads in the retained manuscript summary; hardware handoff used this circuit.                                                                                  |
| Historical cached reduced-surrogate VQE | <code>EfficientSU2</code> ; 152 initial parameters on the 19-flow surrogate. | <code>COBYLA</code> through <code>QiskitOpt.VQE</code> ; 128 optimizer reads; 25 iterations; seeds 74001, 74007, and 74018 in the final follow-up.       | Historical sampled distributions retained in <code>ds_mfg_vqe_reduced_flow_objective_final/vqe_reduced_top50_sampling_summary.csv</code> ; historical optimized ansatz vectors were not retained. | Seed 74018 had 20 reference hits / 524288 reads, 152 top-10 reads, and 391 top-50 reads. The same historical three-seed final-follow-up artifact sums to 28 reference hits / 1572864 reads. |
| Current VQE meta-data/provenance rerun  | <code>EfficientSU2</code> ; 152 initial parameters on the 19-flow surrogate. | <code>COBYLA</code> through <code>QiskitOpt.VQE</code> ; 128 optimizer reads; 25 iterations; seeds 74001, 74007, and 74018 at the final sampling budget. | Current metadata rerun persists optimized ansatz vectors in <code>ds_mfg_vqe_reduced_flow_objective_selected_metadata/</code> .                                                                   | 4 reference hits / 1572864 reads; used for provenance only and not to replace the historical VQE bar in the main manuscript figure or primary-outcomes table.                               |

Table 7: Hardware pilot provenance retained in the submitted-job archive.

| Item             | Retained metadata                                                                                                                                                                         | Limitation                                                                                                    |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Backend and jobs | <code>ibm_fez</code> ; backend queried 2026-06-17 20:19 UTC; 156 qubits; 9 completed jobs; 4096 shots per job; repeats 1–3; transpile seeds 92001–92003; job IDs retained.                | Queue time, calibration snapshot, readout mitigation, and error mitigation metadata were not retained.        |
| Logical circuit  | 19 qubits and 19 classical bits; pre-transpile depth 86; operation counts: 19 h, 95 rx, 95 rz, 255 rzz, and 19 measurements; scoring reverses Qiskit count keys to $x_1, \dots, x_{19}$ . | Submitted-job transpiled depth, two-qubit-gate count, physical layout, and measurement map were not retained. |
| Hardware outcome | 36864 total reads; 6 top-50 reads; 1 top-10 read; 4 stored Gurobi-pool reads; 0 reference-optimum reads; best repaired flow rank 2 with objective 11.8105.                                | The pilot validates submission and scoring only; it cannot diagnose a specific noise mechanism.               |